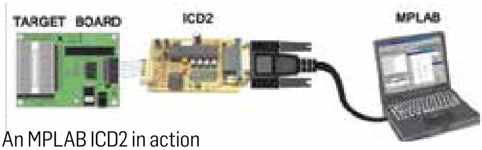
on standard production chips. This greatly decreases the cost associated with production of ICEs, though the name ICE doesn't fit the bill and is actually a misnomer as there is not emulation taking place in this case.

In Circuit Debugger (ICD)

An In Circuit Debugger on the other hand uses special debug hardware added to the target chips which try to provide the developer with ICE-like capabilities. The advantage that ICDs offer is that the code runs on the target chip and not something that emulates the target chip. The cons of using the ICDs are firstly, they require an access to debugging lines which often have multiple roles. You can't use those pins in other roles during a debugging session. ICD also involves adding extra circuitry to the chip, and as the size and cost are a limit many compromises are made on the features available with a chip for eg. tracing which will require a larger RAM for buffering which may not be available. Though these are not much of a problem on larger chips were a few lines dedicated for debugging is not much of an issue but it can severely undermine the performance in a smaller chip.



An MPLAB ICD2 in action

MPLAB ICD3

Emulators

Emulators abstract all the aspects of the hardware and provide them to the user in the development environment. It give the developer the full control of the hardware. This does away with the cost involved in using hardware debuggers but the major downside is that of the speed of operation, the system can sometime be 100x slower than the actual processor.

Tracing

This may as well considered to be a part of debugging but tracing is much different from traditional start and stop debugging thus we're discussing the topic separately. The problem with traditional start/stop debugging technique is that it will not work for a system with real-time constraints. A debugger actually stops the processor as soon as it hits the breakpoint, this